

AMA2111 Mathematics I
2 Linear Algebra

- §1 Matrix (AMA1120)
- §2 System of Linear Equations (AMA1120)
- §3 Vector Space
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§1 Matrix

Definition

- 1 A Matrix is a rectangular array of numbers called scalars, and each scalar in the array is called an entry of the matrix. A matrix A is composed of a finite number of rows and columns. If A has m rows and n columns, then $A_{m \times n} = [a_{ij}]_{m \times n}$ is called an $m \times n$ matrix. The entry at the i -th row and the j -th column of A is denoted by a_{ij} .
- 2 A $1 \times n$ matrix is called a row vector with n entries. The set of all such row vectors is denoted by $\mathbb{R}^n$.
- 3 An $n \times 1$ matrix is called a column vector with n entries. The set of all such column vectors is denoted by $\mathbb{R}_n$.

$$A = \begin{bmatrix} 20.5 & 22 & 22 & 23 \\ 20 & 21 & 21.5 & 22 \\ 19.5 & 20 & 21 & 21.5 \end{bmatrix}, \quad \mathbf{b} = [36 \quad 8], \quad \mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Special Matrix

- 1 Zero Matrix: A matrix with all its entries equal to zero and is denoted by the symbol 0 .

$$0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- 2 Square Matrix: A matrix with n rows and n columns is called an $n \times n$ matrix or a square matrix of order n .

$$A = \begin{bmatrix} 22 & 22 & 23 \\ 20 & 21 & 22 \\ 20 & 21 & 21.5 \end{bmatrix}$$

- 3 Upper Triangular Matrix: A square matrix with all the entries below the main diagonal are zeros.

$$U = \begin{bmatrix} 22 & 22 & 23 \\ 0 & 21 & 22 \\ 0 & 0 & 21.5 \end{bmatrix}$$

Special Matrix

- ④ Lower Triangular Matrix: A square matrix with all the entries above the main diagonal are zeros.

$$L = \begin{bmatrix} 22 & 0 & 0 \\ 20 & 22 & 0 \\ 20 & 21 & 21.5 \end{bmatrix}$$

- ⑤ Diagonal Matrix: A square matrix with $a_{ij} = 0$ whenever $i \neq j$.

$$D = \begin{bmatrix} 22 & 0 & 0 \\ 0 & 21 & 0 \\ 0 & 0 & 21.5 \end{bmatrix}$$

- ⑥ Identity Matrix: A square matrix with $a_{ij} = 0$ whenever $i \neq j$ and $a_{ii} = 1$.

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix Operation

- ① Comparison: Two $m \times n$ matrices A and B are said to be equal if and only if $a_{ij} = b_{ij}$ for $1 \leq i \leq m$ and $1 \leq j \leq n$

$$\begin{bmatrix} 20.5 & 22 & 22 & 23 \\ 20 & 21 & 21.5 & 22 \end{bmatrix} \neq \begin{bmatrix} 20.5 & 22 & 22 \\ 20 & 21 & 21.5 \end{bmatrix}$$

- ② Addition and Subtraction: If A and B are $m \times n$ matrices, we define $A \pm B = [a_{ij} \pm b_{ij}]_{m \times n}$.

$$\begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix} \\ = \begin{bmatrix} 4+1 & 0+1 & 5+1 \\ -1+3 & 3+5 & 2+7 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$$

- ③ Scalar Multiplication: If A is an $m \times n$ matrix and t is any scalar, we define $tA = [ta_{ij}]_{m \times n}$.

$$2 \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 \times 1 & 2 \times 1 & 2 \times 1 \\ 2 \times 3 & 2 \times 5 & 2 \times 7 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix}$$

Matrix Operation

- ④ Matrix Multiplication: If A is an $m \times n$ matrix and B is an $n \times k$ matrix, then we define their product AB to be the $m \times k$ matrix $C_{m \times k} = A_{m \times n} B_{n \times k}$ such that $c_{ij} = \sum_{s=1}^n a_{is} b_{sj}$ for $1 \leq i \leq m$ and $1 \leq j \leq k$.

Example 1

$$A_{2 \times 4} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}, \quad B_{4 \times 3} = \begin{bmatrix} 9 & 13 & 17 \\ 10 & 14 & 18 \\ 11 & 15 & 19 \\ 12 & 16 & 20 \end{bmatrix}$$

$$A_{2 \times 4} B_{4 \times 3} = C_{2 \times 3} = [c_{ij}]_{2 \times 3}$$

Rules of Matrix Algebra

- ① $A + 0 = A, A - A = 0$
- ② $A + B = B + A$
- ③ $A + (B + C) = (A + B) + C$
- ④ $t(A + B) = tA + tB, t(AB) = A(tB)$
- ⑤ $(t + s)A = tA + sA$
- ⑥ $(AB)C = A(BC)$
- ⑦ $(A + B)C = AC + BC, A(B + C) = AB + AC$
- ⑧ $A0 = 0, 0B = 0$
- ⑨ $AI = A, IB = B$

Example 2

$$A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix},$$

compute $AB - BA$.

Matrix Operation

Definition

Let A be a square matrix. We define $A^0 = I$ and $A^k = A(A^{k-1})$ for every positive integer k .

$$A^1 = A, \quad A^2 = AA, \quad A^3 = AAA, \quad \dots$$

Example 3

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Find A^n .

Transpose

Definition

The transpose of an $m \times n$ matrix A is defined to be the $n \times m$ matrix $B = A^T$ such that $b_{ij} = a_{ji}$ for $1 \leq i \leq n$ and $1 \leq j \leq m$.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix}$$

Rules of transpose:

- ❶ $(A^T)^T = A$
- ❷ $(A + B)^T = A^T + B^T$
- ❸ $(kA)^T = kA^T$
- ❹ $(AB)^T = B^T A^T$

Definition

A square matrix A is said to be symmetric if $A = A^T$.

Definition

A square matrix A is said to be nonsingular (or invertible) if there is a square matrix A^{-1} such that $AA^{-1} = I$ or $A^{-1}A = I$. The matrix A^{-1} is called an inverse of A .

$$\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = I \quad \text{and} \quad \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} = I$$

Therefore, the matrix $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ is nonsingular, with $\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$ as an inverse.

Proposition

- ① A^{-1} is unique.
- ② $(A^{-1})^{-1} = A$
- ③ $(tA)^{-1} = \frac{1}{t}A^{-1}$
- ④ If A and B are nonsingular matrices of the same order, then AB is nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.
- ⑤ If A is a nonsingular matrix, $(A^k)^{-1} = (A^{-1})^k$.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}$$

$$AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix} \quad (AB)^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix} = B^{-1}A^{-1}$$

§2 System of Linear Equations

Example

The currents i_1 , i_2 and i_3 in a certain Wheatstone bridge are given by the system of linear equations

$$R_6 i_1 + R_3 (i_1 - i_2) + R_4 (i_1 - i_3) = E_0$$

$$R_3 (i_2 - i_1) + R_2 i_2 + R_5 (i_2 - i_3) = 0$$

$$R_4 (i_3 - i_1) + R_5 (i_3 - i_2) + R_1 i_3 = 0$$

where E_0 is the voltage applied to the circuit and R_1 to R_6 are given resistances. Express currents in terms of resistances and voltage.

The system may be re-written as

$$\begin{bmatrix} R_3 + R_4 + R_6 & -R_3 & -R_4 \\ -R_3 & R_2 + R_3 + R_5 & -R_5 \\ -R_4 & -R_5 & R_1 + R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} E_0 \\ 0 \\ 0 \end{bmatrix}$$

A system of m linear equations with n unknowns $x_1, x_2, \dots, x_n$ is given by

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{cases}$$

where the a_{ij} and the b_k are given scalars. The system can be conveniently written in matrix form as $A\mathbf{x} = \mathbf{b}$ which is followed. The matrix A is commonly known as the coefficient matrix of the linear system.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Row Operation

Definition

Given a system of linear equations $A\mathbf{x} = \mathbf{b}$ we define the augmented matrix for this system to be the $m \times (n + 1)$ matrix obtained by joining to the right of A the column vector $\mathbf{b}$.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

There are 3 Basic Operations:

	Equation	Matrix	Symbol
(i)	Interchange any 2 equations of a system of linear equations	Interchange any two rows of $[\mathbf{A} \mathbf{b}]$	$\mathbf{r}_i \leftrightarrow \mathbf{r}_j$
(ii)	multiply both sides of any equation in a system by a <i>non-zero</i> scalar	multiply any row of $[\mathbf{A} \mathbf{b}]$ by a non-zero scalar	$t \bullet \mathbf{r}_i \rightarrow \mathbf{r}_i$
(iii)	Add a multiple of one equation to another equation within the system	add a scalar multiple of one row of $[\mathbf{A} \mathbf{b}]$ to another row	$t\mathbf{r}_j + \mathbf{r}_i \rightarrow \mathbf{r}_i$

Definition

A matrix is said to be in **row-echelon form** if it has the following properties:

- 1 If a row does not consist entirely of zeros, then the 1st non-zero entry of this row is equal to 1 (known as the leading 1 of the row);
- 2 All the rows that consist entirely of zeros are grouped together at the bottom of the matrix;
- 3 If the leading 1 of the i th row occurs at the p th column and if the leading 1 of the $(i + 1)$ th row occurs at the q th column, then $p < q$.

If each column of a row-echelon form that contains a leading 1 has zeros elsewhere, the matrix is called **reduced row-echelon form**.

Row-echelon Form

$$\begin{bmatrix} 1 & 4 & 3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 18 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Reduced Row-echelon Form

$$\begin{bmatrix} 0 & 1 & 2 & 0 & 15 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -9 \end{bmatrix}$$

Gaussian Elimination

- 1 Use a sequence of elementary row operations to reduce the augmented matrix $[A|\mathbf{b}]$ to a matrix $[R|\mathbf{c}]$, where R is the row-echelon form of A .
- 2 The system of linear equations $R\mathbf{x} = \mathbf{c}$, which is equivalent to $A\mathbf{x} = \mathbf{b}$, may now be solved by backward substitutions.

Example 4

Use Gaussian elimination to solve

$$\begin{cases} x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - 3x_3 = 1 \\ 3x_1 + 6x_2 - 5x_3 = 0 \end{cases}$$

Example 5

Use Gaussian elimination to solve

$$\begin{cases} x_1 + x_2 - 4x_3 = 5 \\ 2x_1 + 3x_2 - 7x_3 = 14 \\ -x_2 - x_3 = -4 \end{cases}$$

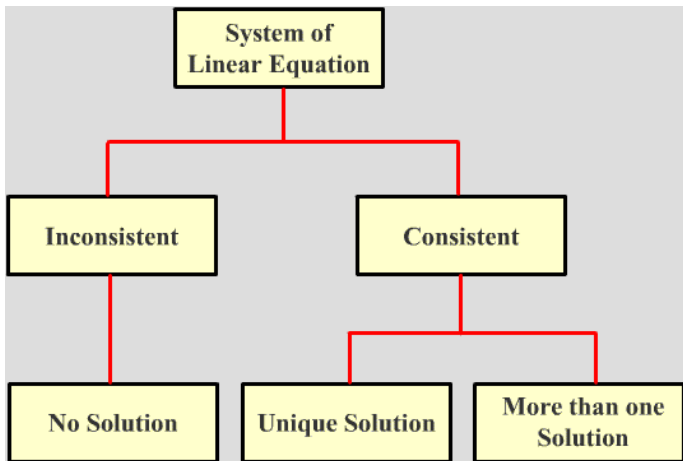
Example 6

Use Gaussian elimination to solve

$$\begin{cases} x_2 - 4x_3 = 8 \\ 2x_1 - 3x_2 + 2x_3 = 1 \\ 5x_1 - 8x_2 + 7x_3 = 1 \end{cases}$$

Solution Set

The collection of all the solutions is called the solution set of the system. A system of linear equations is either inconsistent (no solution) or consistent (has at least one solution).



Theorem

Suppose the augmented matrix $[A|\mathbf{b}]$ of the linear system $A\mathbf{x} = \mathbf{b}$ is reduced to $[R|\mathbf{c}]$ by elementary row operations, where R is an $m \times n$ matrix in the row-echelon form and $\mathbf{c} \in \mathbb{R}_m$.

- 1 If R has $r(< m)$ non-zero rows, the system $A\mathbf{x} = \mathbf{b}$ has no solution if and only if there is at least one $c_j \neq 0$ for $r < j \leq m$. So the system is inconsistent.
- 2 Otherwise the system is consistent.
 - a If R has n non-zero rows, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.
 - b If R has $r(< n)$ non-zero rows, the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions if and only if all $c_j = 0$ for $r < j \leq m$.

Example

- 1 Example 4 has a unique solution, as R has 3 non-zero rows while A is 3×3 matrix. The number of non-zero rows (pivots) of R equals to the number of unknowns, both 3.
- 2 Example 5 has infinitely many solutions, as R has 2 non-zero rows and $c_3 = 0$ while A is 3×3 matrix. The number of pivots of R is less than the number of unknowns, $2 < 3$.
- 3 Example 6 has no solution, as R has 2 non-zero rows and $c_3 \neq 0$ while A is 3×3 matrix. One equation is inconsistent.

Example 7

Use Gaussian elimination to solve

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1 \\ 5x_3 + 10x_4 + 15x_6 = 5 \\ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6 \end{cases}$$

Example 8

Find the conditions which p and q must satisfy for the system

$$\begin{bmatrix} 1 & 2 & p \\ 2 & 3 & 0 \\ 6 & 8 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ q \end{bmatrix}$$

to have (a) a unique solution, (b) infinitely many solutions and (c) no solution.

Example 9

Consider the linear system

$$\begin{bmatrix} 1 & 2 & 4 & 7 \\ 2 & 3 & 4 & 6 \\ 3 & 4 & a & 6 \\ -1 & 0 & 4 & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$$

Find the conditions satisfied by a and b such that the system has

- ❶ no solutions;
- ❷ infinitely many solutions;
- ❸ a unique solution.

Gauss-Jordan Method

- 1 Use a sequence of elementary row operations to reduce the augmented matrix $[A|\mathbf{b}]$ to a matrix $[R|\mathbf{c}]$, where R is the reduced row-echelon form of A .
- 2 The solution of the new equivalent system $R\mathbf{x} = \mathbf{c}$ can be obtained simply by inspection.

Example 10

Use Gauss-Jordan Method to solve

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 6 \\ -4x_1 + 5x_2 + 9x_3 = -9 \end{cases}$$

Use Gauss-Jordan Method to Find Inverse

Given $A_{n \times n}$ is a square matrix, $\mathbf{b}_i (i = 1, 2, \dots, n)$ is the i th column of the identity matrix $I_{n \times n}$. According to Gauss-Jordan method, if $A\mathbf{x}_1 = \mathbf{b}_1$ has a unique solution, $A\mathbf{x}_i = \mathbf{b}_i$ ($i = 2, \dots, n$) has a unique solution too.

Since $A[\mathbf{x}_1 \mathbf{x}_2 \dots \mathbf{x}_n] = [\mathbf{b}_1 \mathbf{b}_2 \dots \mathbf{b}_n] = I$, $A^{-1} = [\mathbf{x}_1 \mathbf{x}_2 \dots \mathbf{x}_n]$. So we can use Gauss-Jordan method to find inverse, i.e. solve $A_{n \times n} X_{n \times n} = I_{n \times n}$ to get $A^{-1} = X$.

$$[A|I] \rightarrow [I|A^{-1}]$$

Example 11

Find the inverse of

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$$

System of Homogeneous Equations

Definition

- 1 A system of m homogeneous linear equations in n unknowns is given by $A\mathbf{x} = \mathbf{0}$, where A is a given $m \times n$ matrix and $\mathbf{0}$ is the $m \times 1$ zero column vector.
- 2 A homogeneous system has the $n \times 1$ zero vector $\mathbf{0}$ as an obvious solution (called the **trivial solution**). Any other solutions are known as non-trivial solutions.

Theorem

The solution set of $A\mathbf{x} = \mathbf{0}$ either has only the trivial solution, or has infinitely many solutions.

§3 Vector Space

Definition

A vector space is a set V of column vectors (or row vectors) satisfying the following properties:

- 1 contains the zero vector $\mathbf{0}$;
- 2 if V contains $\mathbf{v}$ and $\mathbf{w}$, then V also contains the sum $\mathbf{v} + \mathbf{w}$;
- 3 if V contains $\mathbf{v}$ and t is any scalar, then V also contains $t\mathbf{v}$.

A vector space contains the zero vector, and is closed under addition and scalar multiplication.

Example

Both $\mathbb{R}_n$ and $\mathbb{R}^n$ are vector spaces in their own right. In particular, $\mathbb{R}_2$ is the usual Cartesian plane and $\mathbb{R}_3$ is the Cartesian space.

Definition

Let $\mathbf{v}_i \in \mathbb{R}_n$ and $t_i \in \mathbb{R}$, $i = 1, 2, \dots, k$.

- ① A vector $\mathbf{v}_k$ is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}$ if $\mathbf{v}_k = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_{k-1}\mathbf{v}_{k-1}$.
- ② Suppose that W is the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$. Then W is a vector space spanned by the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$, called $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.
- ③ Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are said to be **linearly dependent** if there are scalars t_i , not all 0, such that $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_k\mathbf{v}_k = \mathbf{0}$.
- ④ Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are said to be **linearly independent** if and only if $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_k\mathbf{v}_k = \mathbf{0} \Rightarrow t_i = 0$ for all $i = 1, 2, \dots, k$.

Proposition

- 1 Vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly dependent if and only if one of these vectors can be expressed as a linear combination of the remaining ones.
- 2 A is denoted by the $n \times k$ matrix whose j th column equals to the vector $\mathbf{v}_j$ for $j = 1, \dots, k$. Clearly,

$$t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_k\mathbf{v}_k = \mathbf{0} \Leftrightarrow A\mathbf{t} = \mathbf{0}$$

$$\text{where } \mathbf{t} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_k \end{bmatrix}, A = [\mathbf{v}_1 \mathbf{v}_2 \dots \mathbf{v}_k].$$

Therefore, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly dependent if and only if the system of homogeneous equations $A\mathbf{t} = \mathbf{0}$ has infinitely many solutions; $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent if and only if the system of homogeneous equations $A\mathbf{t} = \mathbf{0}$ has only the trivial solution.

Example 12

Determine $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are linearly independent or dependent.

Example 13

Determine $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$ are linearly independent or dependent. If dependent, could $\mathbf{v}_3$ be expressed by the linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Proposition

Consider $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{R}_n$, $\text{span} \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} = \mathbb{R}_n$ if and only if there are n linearly independent vectors in $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$.

Example 14

Determine $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$ can span $\mathbb{R}_3$ or not.

Definition

The **rank** of a matrix A is the dimension of the vector space spanned by its columns. This corresponds to the maximal number of linearly independent columns of A .

- 1 Refer to Example 4, the coefficient matrix A has 3 linearly independent columns. $\text{Rank}(A) = \# \text{pivots} = 3$.
- 2 Refer to Example 5, the coefficient matrix A has 2 linearly independent columns. $\text{Rank}(A) = \# \text{pivots} = 2$.

§4 Determinant

Definition

For any square matrix $A_{n \times n}$, we define the **determinant** of A , denoted by $|A|$ or $\det(A)$, as follows:

- If $n = 1$, i.e., $A = [a_{11}]$, we define $|A| = a_{11}$.
- Assume that $n > 1$ and that the determinant is defined for all square matrices of order $< n$. Let $A_{n \times n} = [a_{ik}]$, we define the following terms:
 - ① M_{ik} is the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by deleting its i th row and k th column. M_{ik} is called the minor of the entry a_{ik} of A .
 - ② $C_{ik} = (-1)^{i+k} M_{ik}$ is called the cofactor of the entry a_{ik} of A .

$$|A| = \sum_{k=1}^n a_{1k} C_{1k}$$

$|A|$ is obtained by taking “cofactor expansion” along the first row of the matrix A .

Example

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$|A| = a_{11}C_{11} + a_{12}C_{12} = a_{11}a_{22} - a_{12}a_{21}$$

Example 15

Evaluate $|A|$ of

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$$

Theorem

- 1 The determinant of a matrix can be evaluated by taking cofactor expansion along any row.
- 2 The determinant of a matrix can be evaluated by taking cofactor expansion along any column.

The sign pattern of cofactor is

$$\begin{bmatrix} + & - & + & - & & \\ - & + & - & + & & \\ + & - & + & - & \dots & \\ - & + & - & + & & \\ & & \vdots & & & \end{bmatrix}$$

Calculate the determinant of the matrix in Example 15 by expanding along the third row and obtain

$$|A| = 1 \cdot \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} + 4 \cdot (-1) \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} + (-2) \cdot \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = 14$$

Proposition

- ① $|A^T| = |A|$
- ② If A' is the matrix obtained from A by interchanging any two rows of A , then $|A'| = -|A|$.
- ③ If A' is the matrix obtained by multiplying the i th row of A by a scalar t while other rows remain unchanged, then $|A'| = t|A|$.
- ④ If A' is obtained from A by an elementary row operation of adding a scalar multiple of the i th row to its j th row, where $i \neq j$, then $|A'| = |A|$.
- ⑤ If two rows of A are identical, then $|A| = 0$.

⑥

$$D = \begin{bmatrix} d_{11} & 0 & \dots & 0 \\ 0 & d_{22} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & d_{nn} \end{bmatrix} \quad U = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}$$
$$|D| = d_{11}d_{22} \dots d_{nn} \quad |U| = a_{11}a_{22} \dots a_{nn}$$

Proposition

7 A square matrix A is nonsingular if and only if $|A| \neq 0$.

8

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} + b_1 & a_{i2} + b_2 & \cdots & a_{in} + b_n \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \\ = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ b_1 & b_2 & \cdots & b_n \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

9 If A and B are $n \times n$ matrices, then $|AB| = |A||B|$.

Example 16

Evaluate

$$\begin{vmatrix} 0 & -1 & 2 & 1 \\ -4 & 3 & -3 & 5 \\ 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 1 \end{vmatrix}$$

Theorem

For any square matrix $A_{n \times n}$, the following statements are equivalent.

- 1 The system of non-homogeneous equations $A\mathbf{x} = \mathbf{b}$ has a unique solution;
- 2 A can be reduced to I by a sequence of elementary row operations;
- 3 A is nonsingular;
- 4 The system of homogeneous equations $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
- 5 A is denoted by the $n \times n$ matrix whose j th column equals to the vector $\mathbf{v}_j$ for $j = 1, \dots, n$, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.
- 6 $\text{Rank}(A) = n$.
- 7 $|A| \neq 0$;

§5 Eigenvalue and Eigenvector

Definition

Let A be an $n \times n$ matrix. A non-zero vector $\mathbf{v} \in \mathbb{R}_n$ is called an **eigenvector** of A if $A\mathbf{v}$ is a scalar multiple of $\mathbf{v}$, i.e., if there is a scalar λ such that $A\mathbf{v} = \lambda\mathbf{v}$. The scalar λ is called an **eigenvalue** of A and $\mathbf{v}$ is said to be an eigenvector of A corresponding to the eigenvalue λ .

If $\mathbf{v}$ is an eigenvector of A corresponding to the eigenvalue λ , then $t\mathbf{v}$ is also an eigenvector of A corresponding to the same eigenvalue λ , provided that $t \neq 0$.

Theorem

Let A be an $n \times n$ matrix with real entries. A real number λ is an eigenvalue of A if and only if $|A - \lambda I| = 0$.

Proof:

By rewriting $A\mathbf{v} = \lambda\mathbf{v}$ as $A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$ or $(A - \lambda I)\mathbf{v} = \mathbf{0}$, we observe that finding an eigenvector of A is equivalent to finding non-trivial solutions of the homogeneous system $(A - \lambda I)\mathbf{v} = \mathbf{0}$. We conclude that $(A - \lambda I)\mathbf{v} = \mathbf{0}$ has non-trivial solutions if and only if $|A - \lambda I| = 0$.

Characteristic Polynomial

Definition

$$f(\lambda) = |A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix}$$

$f(\lambda)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

$f(\lambda)$ is called the **characteristic polynomial** of A .

Therefore, eigenvalues of the matrix A are roots of the equation $f(\lambda) = 0$. As a result, an $n \times n$ matrix has at most n real eigenvalues.

Once we obtain an eigenvalue λ of A , we shall be able to use Gaussian elimination to find non-trivial solutions of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ and thus obtain the corresponding eigenvectors of A .

As the homogeneous system $(A - \lambda I)\mathbf{v} = \mathbf{0}$ has infinitely many non-trivial solutions, there are infinitely many eigenvectors of A corresponding to λ . Therefore, we only need to find eigenvectors that are linearly independent. All other eigenvectors corresponding to λ may be expressed as linear combinations of these linearly independent eigenvectors.

Example 17

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$$

Example 18

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Example 19

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & -1 \\ -1 & 2 & 0 \end{bmatrix}$$

Definition

A square matrix A is said to be **diagonalizable** if there is a nonsingular matrix P such that $P^{-1}AP$ is a diagonal matrix. We also say that the matrix P diagonalizes A .

P is not unique.

Theorem

Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent eigenvectors of A corresponding to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (not necessarily distinct), then by taking P the $n \times n$ matrix having $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ as its columns and D to be the diagonal matrix with $d_{jj} = \lambda_j$ for $j = 1, 2, \dots, n$, we obtain $AP = PD$.

Both Example 17 and 18 have 3 linearly independent eigenvectors, so they are diagonalizable.

For Example 17,

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

For Example 18,

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

But Example 19 only has 2 linearly independent eigenvectors, so it is not diagonalizable.

Proposition

If A is diagonalizable and $A = PDP^{-1}$, then $A^m = PD^mP^{-1}$,

$$\text{where } D^m = \begin{bmatrix} \lambda_1^m & 0 & \cdots & 0 \\ 0 & \lambda_2^m & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^m \end{bmatrix}.$$

Proof:

$$\begin{aligned} A^m &= PDP^{-1}PDP^{-1}P \cdots P^{-1}PDP^{-1} = PDIDI \cdots IDP^{-1} \\ &= PDD \cdots DP^{-1} = PD^mP^{-1} \end{aligned}$$

This method is important in control theory.

Example 20

Compute A^{290} , where $A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

Example 21

Diagonalize $2A + 3I$ if P diagonalizes A .

§6 Inner Product and Orthogonality

Definition

If $\mathbf{v}$ and $\mathbf{w}$ are vectors in $\mathbb{R}_n$, we define the **inner product** of $\mathbf{v}$ and $\mathbf{w}$

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w} = v_1 w_1 + \dots + v_n w_n,$$

where $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are entries of the vectors $\mathbf{v}$ and $\mathbf{w}$, respectively.

Proposition

- ① $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for any $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}_n$.
- ② $\langle \mathbf{v} + \mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ for any $\mathbf{v}, \mathbf{u}$ and $\mathbf{w}$ in $\mathbb{R}_n$.
- ③ $\langle t\mathbf{v}, \mathbf{w} \rangle = t\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, t\mathbf{w} \rangle$ for any $\mathbf{v}, \mathbf{w}$ in $\mathbb{R}_n$ and for any scalar t .
- ④ $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ for any $\mathbf{v}$ in $\mathbb{R}_n$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Definition

- ① The **Euclidean norm** of a vector $\mathbf{v}$ is defined by

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

- ② If $\|\mathbf{u}\| = 1$, then $\mathbf{u}$ is said to be a **unit vector**.

Normalization

If $\mathbf{v}$ is a non-zero vector, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a unit vector.

Proposition

The angle between $\mathbf{u}$ and $\mathbf{v}$ is the unique number θ lying between 0 and π such that $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$.

Example

Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be nonzero vectors. We observe that

$$\begin{aligned}\|\mathbf{v} - \mathbf{u}\|^2 &= (v_1 - u_1)^2 + (v_2 - u_2)^2 \\ &= v_1^2 - 2u_1v_1 + u_1^2 + v_2^2 - 2u_2v_2 + u_2^2 \\ &= v_1^2 + v_2^2 + u_1^2 + u_2^2 - 2(u_1v_1 + u_2v_2) \\ &= \|\mathbf{v}\|^2 + \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle\end{aligned}$$

From the law of cosines, we have

$$\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$

Hence,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$

Orthogonality

Definition

- 1 Two vectors $\mathbf{v}$ and $\mathbf{w}$ are said to be **orthogonal** if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.
- 2 A set of vectors $v_1, \dots, v_k$ is said to be an orthogonal set of vectors if these vectors are mutually orthogonal to each other, i.e., $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ whenever $i \neq j$.
- 3 An orthogonal set of vectors $v_1, \dots, v_k$ is said to be orthonormal if all these vectors are unit vectors, i.e., $\|\mathbf{v}_i\| = 1$ for all i .

Example 22

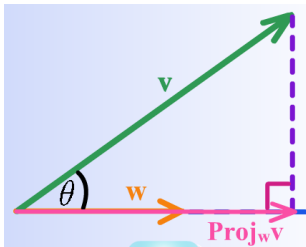
Is $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$ an orthogonal set of vectors? Can it be normalized to an orthonormal set of vectors?

Gram-Schmidt Process

Definition

Let $\mathbf{v}$ and $\mathbf{w}$ be vectors and $\mathbf{w} \neq \mathbf{0}$. We define the projection of $\mathbf{v}$ onto $\mathbf{w}$ to be the vector

$$\text{proj}_{\mathbf{w}} \mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\|\mathbf{w}\|^2} \mathbf{w}$$



$\mathbf{v} - \text{proj}_{\mathbf{w}} \mathbf{v}$ is orthogonal to $\mathbf{w}$. This can be applied to a construction of orthonormal vectors known as the Gram-Schmidt process.

Example 23

Use the Gram-Schmidt process to transform the given vectors into an orthonormal set of vectors.

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Example 24 (*)

Use the Gram-Schmidt process to transform the given vectors into an orthonormal set of vectors.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Orthogonal Matrix

Definition

A square matrix Q is called an orthogonal matrix if $Q^{-1} = Q^T$, i.e., $QQ^T = Q^TQ = I$.

If $Q = [\mathbf{q}_1 \ \dots \ \mathbf{q}_n]$, then $\{\mathbf{q}_1 \ \dots \ \mathbf{q}_n\}$ is an orthonormal set of vectors, i.e.,

- 1 $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = 0$ whenever $i \neq j$.
- 2 all these $\mathbf{q}_i$ are unit vectors.

Theorem

Let A be a symmetric matrix of order n . Then

- 1 A has eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ which form an orthogonal matrix $Q = [\mathbf{q}_1 \ \dots \ \mathbf{q}_n]$;
- 2 there is an orthogonal matrix Q such that $A = QDQ^T$, where D is a diagonal matrix.

Example 25

Let $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.

Find an orthogonal matrix Q such that $Q^T A Q$ is a diagonal matrix.